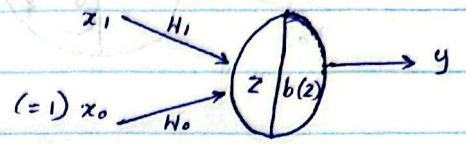
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Train a model to learn the AND gate

A logic gate produces an output from binary inputs. (0 or 1)

An artificial neuron can be trained using the input-output examples of a logic gate



For two inputs : x_1 , x_2

The neuron calculates : $z = H_0 + H_1 x_1 + H_2 x_2$

Produces : $\hat{y} = f(z)$; Where f is the activation function.

AND gate

Features		target
x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1

The neuron adjusts its weights until its outputs are close to these target values.

Activation Functions

An activation function determines how the weighted input z is converted into neuron output.

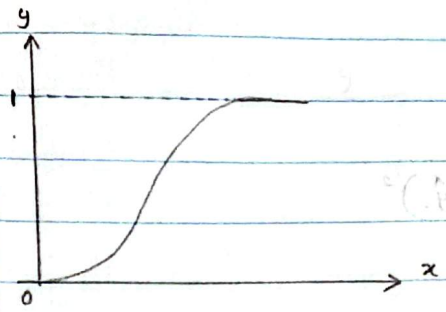
$$\hat{y} = f(z)$$

There are 4 important activation functions.

- Sigmoid
- Tanh
- Linear activation
- ReLU

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Sigmoid

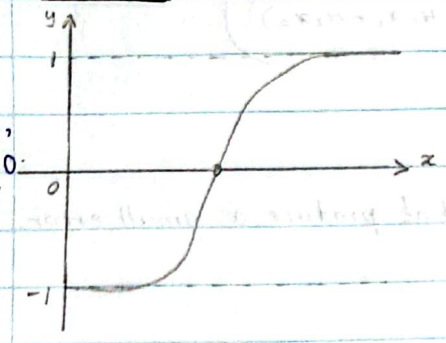


* 0 to 1
* S-shaped

This is suitable when the required output is between 0 and 1.

$$\sigma(z) = \frac{1}{1+e^{-z}} \quad ; \quad 0 < \sigma(z) < 1$$

Tanh

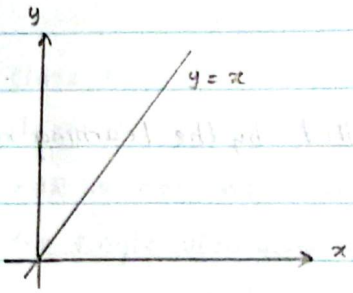


* -1 to 1
* S-shaped, centered at 0.

The hyperbolic tangent function produces values between -1 and 1.

$$f(z) = \tanh(z) \quad ; \quad -1 < f(z) < 1$$

Linear Activation

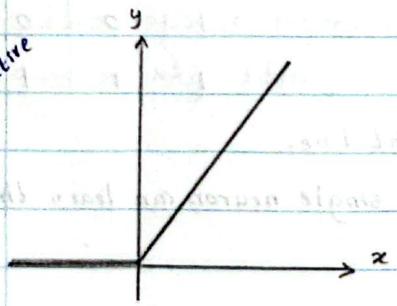


* $-\infty$ to $+\infty$
* Straight line.

This creates a linear relationship between input & output.

$$f(z) = z$$

ReLU



* 0 to $+\infty$
* 0 for negative input.

ReLU stands for Rectified Linear Unit.

$$f(z) = \max(0, z)$$

$$f(z) = \begin{cases} 0 & , \ z < 0 \\ z & , \ z \geq 0 \end{cases}$$

* Negative values become 0, while positive values remain positive.

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Error during the AND gate learning

For training sample i : $e_i = y_i - \hat{y}_i$

Using squared error : $E = \sum_{i=1}^{f} (y_i - \hat{y}_i)^2$

$$E = \sum_{i=1}^{f} \left(y_i - \frac{1}{1+e^{-z}}\right)^2$$

$$E = \sum_{i=1}^{4} \left(y_i - \frac{1}{1+e^{-(W_0 x_0 + W_1 x_1 + W_2 x_2)}}\right)$$

$$E = f(W_1, W_2, W_0)$$

The aim of training is to find weights that produce a small error.

Weight Adjustment

The derivative of the error shows how strongly the error changes when a weight changes.

$$\frac{\partial E}{\partial W_0}$$

The size of the weight adjustment is controlled by the learning rate: η

$$\Delta W_0 = \eta \cdot \frac{\partial E}{\partial W_0}$$

0	0	0
0	1	0
1	0	0
1	1	1

* AND can be learned by one neuron.
  - class 0 : (0,0), (0,1), (1,0)
  - class 1 : (1,1)

These points can be seperated by one straight line.

Therefore AND is linearly separable. & a single neuron can learn the AND gate.

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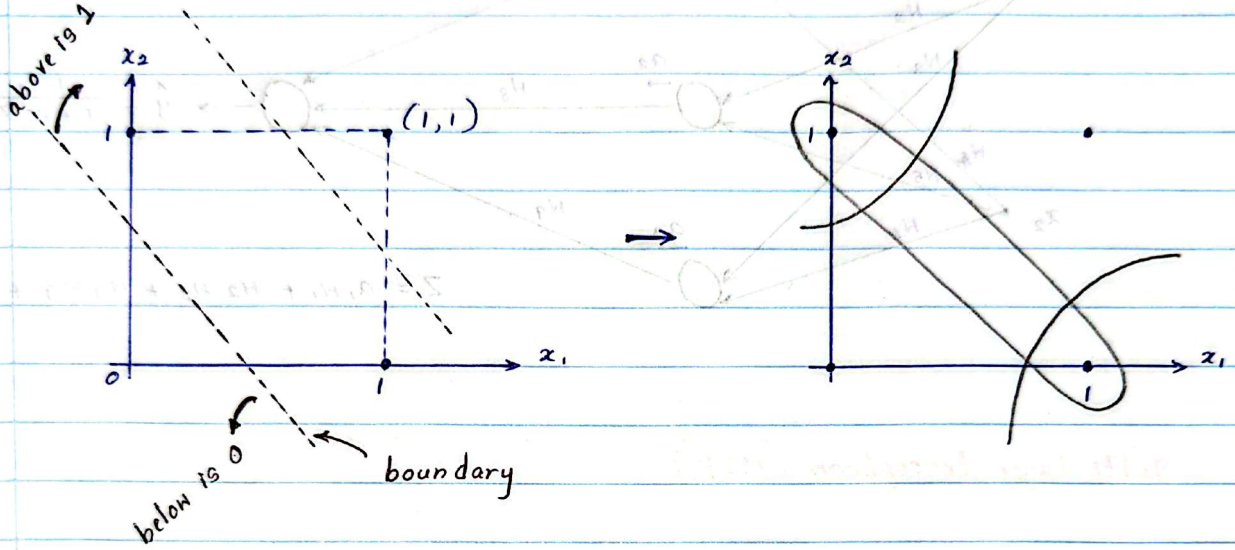
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XOR gate

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0

0.03 → 0
0.97 → 1
0.96 → 1
0.04 → 0



class 1 : (0,1), (1,0)
class 0 : (0,0), (1,1)

* XOR is not ~~Lani~~ linearly separable.
* No single straight line can place both class 1 points on one side and both class 0 points on the other side.
* So single neuron cannot correctly solve XOR.

* Basic logistic regression also cannot directly solve XOR.
* This is why adding more neurons become necessary.

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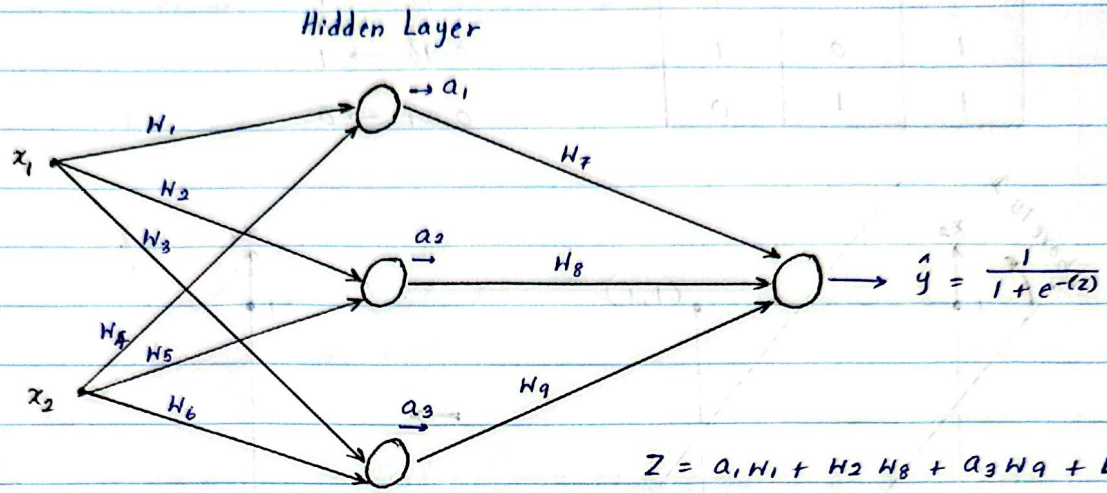
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Hidden Layer

A hidden layer is an intermediate layer located between the input layer and output layer.



Multi Layer Perceptron (MLP)

MLP is a neural network that contains an input layer, at least one hidden layer and an output layer.

+ Each hidden neuron can represent a different boundry or intermediate pattern.

One neuron ⟶ one linear boundry.

Hidden neuron 1 ⟶ boundary 1
Hidden neuron 2 ⟶ boundary 2

+ Hidden layers allow a neural network to solve XOR.

+ Activation functions introduce nonlinear behaviour into the network.

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